

Example 4: Solve: $81 \div 9 + 34$

Solution: As division and addition both are involved in question, we have to solve the division first then addition.

$$\begin{aligned} 81 \div 9 + 34 &= 81 \div 9 + 34 \\ &= 9 + 34 = 43 \end{aligned}$$

EXERCISE 1.13

Solve the following:

(1) $46 - 23 + 17$

(3) $98 - 46 + 24$

(5) $324 - 152 + 182$

(7) $682 + 329 - 159$

(9) $253 \div 11 \times 5$

(11) $992 \div 16 \times 4$

(13) $12 \times 114 \div 19 + 10$

(15) $32 \times 400 \div 16 + 23$

(2) $99 - 77 + 33$

(4) $48 - 21 + 31$

(6) $582 - 325 + 154$

(8) $489 \div 5 + 393$

(10) $540 \div 15 \times 8$

(12) $7 \times 375 \div 15$

(14) $23 + 800 \div 20 \times 2$

(16) $451 \times 690 \div 30 - 15$

Solve real life problems (using Pakistani currency as well) involving addition, subtraction, multiplication and division.

Addition, Subtraction, Multiplication and Division:

Example 1: Sara spent Rs 486,935 to buy a car and Rs 439,870 to buy Jewellery. How much money she spent altogether?

Solution: To buy a car	= Rs	4 8 6 9 3 5
To buy Jewellery	= Rs	+ 4 3 9 8 7 0
Total amount spent by Sara.	= Rs	<u>9 2 6 8 0 5</u>

Example 2: There were 767,513 people in a stadium to watch a football match. After break only 468,302 people remained in the stadium. How many people left the stadium?

Solution: Number of people came
remaining people

Number of people left the stadium

$$\begin{array}{r}
 \overset{6}{7} \overset{10}{6} \overset{10}{7} \overset{10}{5} \overset{10}{1} \overset{10}{3} \\
 - 4 \overset{10}{6} \overset{10}{8} \overset{10}{3} \overset{10}{0} \overset{10}{2} \\
 \hline
 2 \overset{10}{9} \overset{10}{9} \overset{10}{2} \overset{10}{1} \overset{10}{1}
 \end{array}$$

Example 3: A company sold 856,940 fans in summer season and earned Rs 341 profit on each fan. Find the total profit.

Solution:

Number of fans
Profit on each fan

$$\begin{array}{r}
 \overset{1}{8} \overset{2}{5} \overset{2}{6} \overset{1}{9} \overset{1}{4} \overset{1}{0} \\
 \times \quad \overset{1}{3} \overset{1}{4} \overset{1}{1} \\
 \hline
 \overset{1}{3} \overset{1}{4} \overset{2}{2} \overset{1}{7} \overset{1}{7} \overset{1}{6} \overset{1}{0} \overset{1}{x} \\
 2 \overset{1}{5} \overset{1}{7} \overset{1}{0} \overset{1}{8} \overset{1}{2} \overset{1}{0} \overset{1}{0} \\
 \hline
 2 \overset{1}{9} \overset{1}{2} \overset{1}{2} \overset{1}{1} \overset{1}{6} \overset{1}{5} \overset{1}{4} \overset{1}{0}
 \end{array}$$

Hence the company had profit of Rs 292,216,540.

Example 4: A shopkeeper sold 3,104 note books in 16 weeks. He sold exact number of note books in each week. How many note books sold in a week?

Solution:

Number of note books sold = 3104
Number of weeks = 16

$$\begin{array}{r}
 16 \overline{) 3104} \\
 \underline{- 16} \\
 150 \\
 \underline{- 144} \\
 64 \\
 \underline{- 64} \\
 00
 \end{array}$$

Hence, 194 note books were sold in a week.

EXERCISE 1.14

- 1 Cost of a car is Rs 748,630 and the cost of another car is Rs 630,010. What is the total cost of both cars?
- 2 Shagufta had Rs 389,000 in her bank account. She spent Rs 183,499 for repair of her house. How much amount is left with her?
- 3 A poultry farm sold 143,860 chicken in a month and 354,180 in next month. Find the difference in its sale.
- 4 The cost of a TV is Rs 95,400. Find the cost of such 150 Tvs.
- 5 A farm contains 56,321 trees of dates. How many date trees will be in 835 farms?
- 6 A school spent Rs 4,375 on a picnic party. Only 35 students went to a picnic. What is the amount to be paid by each student?
- 7 Najeeb is distributing 3,290 food packs among 235 families. How many food packs will each family get?

REVIEW EXERCISE – 1

1. **Tick the correct options.**
 - (i) The place value of 8 in 2,485,612 is _____
(a) 800 (b) 8,000 (c) 80,000
(d) 800,000
 - (ii) Seven digit numbers represent _____
(a) thousand (b) millions (c) ten millions
(d) hundred millions
 - (iii) $20 + 8 \div 4 =$ _____
(a) 4 (b) 7 (c) 8 (d) 22
2. **Write the following numbers in words.**
(i) 2,412,316 (ii) 36,123,101 (iii) 600,216
3. **Write the following numbers in figures.**
(i) Twenty million (ii) One hundred million
4. **Add.**
(i) 416,712 and 712,145 (ii) 900,102 and 812,156
5. **Subtract.**
(i) 218,822 from 967,829 (ii) 100,512 from 200,603
(iii) 555,666 from 723,444
6. **Solve the following.**
(i) 61243×261 (ii) 21588×120
(iii) $3810 \div 15$ (iv) $5088 \div 32$
(v) $555 \div 15 \times 2$
7. The annual saving of Raheel is Rs 89,560. Out of which he purchases a T.V of Rs 35,000. How much amount is left with him?

FACTORS AND MULTIPLES

2.1 DIVISIBILITY TESTS

Identify divisibility rules for 2, 3, 5 and 10

A divisibility test is a quick way of testing if a given number is divisible by another number without doing the division.



Activity

Find the factors of the numbers

2, 4, 5, 7, 9, 12, 18, 24, 32 and 48.

Every number is divisible by 1

Let us examine the numbers.

Example 1: Which of the following numbers is divisible by 2?

(i) 1,456

Digit at units place is 6, which is even.

So, 1,456 is divisible by 2

Verification:

$$\begin{array}{r}
 728 \\
 2 \overline{) 1456} \\
 \underline{-14} \\
 5 \\
 \underline{-4} \\
 16 \\
 \underline{-16} \\
 00
 \end{array}$$

(ii) 92,357

Digit at units place is 7, which is odd.

So, 92,357 is not divisible by 2

Verification:

$$\begin{array}{r}
 46268 \\
 2 \overline{) 92537} \\
 \underline{-8} \\
 12 \\
 \underline{-12} \\
 5 \\
 \underline{-4} \\
 13 \\
 \underline{-12} \\
 17 \\
 \underline{-16} \\
 1
 \end{array}$$

Note: Remainder is not 0

So, 92537 is not exactly divisible by 2.

A number is divisible by 2, if ones digit is 0, 2, 4, 6 or 8

Example 2: Which of the following numbers is divisible by 3?

- (i) 1,452 (ii) 62,345

Solution:

(i) 1,452

Sum of digit is $1 + 4 + 5 + 2 = 12$

and 12 is divisible by 3

So, 1,452 is also divisible by 3

Verification:

$$\begin{array}{r} 3 \overline{) 1452} \quad (484 \\ \underline{-12} \\ 25 \\ \underline{-24} \\ 12 \\ \underline{-12} \\ 00 \end{array}$$

(ii) 62,345

Sum of digits $6 + 2 + 3 + 4 + 5 = 20$

and 20 is not divisible by 3

So, 62,345 is not divisible by 3

Verification:

$$\begin{array}{r} 3 \overline{) 62345} \quad (20748 \\ \underline{-6} \\ 23 \\ \underline{-21} \\ 14 \\ \underline{-12} \\ 25 \\ \underline{-24} \\ 1 \end{array}$$

Note: Remainder is not 0

So, 62,345 is not divisible by 3.

Add the digits. If the result is divisible by 3, then the original number is also divisible by 3.

Example 3: Which of the following numbers is divisible by 5?

(i) 14673

14673 is not divisible by 5 because digit at unit place is not 0 or 5

Verification:

$$\begin{array}{r} 2934 \\ 5 \overline{) 14673} \\ \underline{-10} \\ 46 \\ \underline{-45} \\ 17 \\ \underline{-15} \\ 23 \\ \underline{-20} \\ 3 \end{array}$$

Note: Remainder is not 0

So, 14,673 is not divisible by 5.

(ii) 31360

31360 is divisible by 5 because digit at ones place is 0

Verification:

$$\begin{array}{r} 6272 \\ 5 \overline{) 31360} \\ \underline{-30} \\ 13 \\ \underline{-10} \\ 36 \\ \underline{-35} \\ 10 \\ \underline{-10} \\ 0 \end{array}$$

A number is divisible by 5, if the last digit is 0 or 5.

Example 4: Which of the following number is divisible by 10?

- (i) 24,563 (ii) 16,230

Solution:

(i) 24,563

24,563 is not divisible by 10 because digit at ones place is not zero

Verification:

$$\begin{array}{r}
 10 \overline{) 24563} \quad (2456 \\
 \underline{- 20} \\
 45 \\
 \underline{- 40} \\
 56 \\
 \underline{- 50} \\
 63 \\
 \underline{- 60} \\
 3
 \end{array}$$

Note: Remainder is not 0

So, 24,563 is not divisible by 10.

(ii) 16,230

16,230 is divisible by 10 because digit at ones place is zero.

Verification:

$$\begin{array}{r}
 10 \overline{) 16230} \quad (1623 \\
 \underline{- 10} \\
 62 \\
 \underline{- 60} \\
 23 \\
 \underline{- 20} \\
 30 \\
 \underline{- 30} \\
 00
 \end{array}$$

A number is divisible by 10, if last digit is 0.

Use divisibility tests for 2, 3, 5 and 10 on numbers up to 5-digits



Activity

Check the divisibility of 15,381 by 2, 3, 5 and 10.

Solution:

- (1) 15,381 since the digit at the unit place is 1, therefore, 15,381 is not divisible by 2.
- (2) The sum of the digits in 15,381 is: $1 + 5 + 3 + 8 + 1 = 18$ as 18 is divisible by 3; therefore 15,381 is divisible by 3.
- (3) In 15,381, the digits at ones place is not 0 or 5; therefore, 15,381 is not divisible by 5.
- (4) Also the digit at ones place is not '0'; therefore, 15,381 is not divisible by 10.

Teacher's Note

Teacher should do enough practice of divisibility rules with the help of examples.

EXERCISE 2.1

1. Which of the following numbers are divisible by 2?

(i) 120 (ii) 1,001 (iii) 1,434 (iv) 2,221 (v) 13,574

2. Test the following numbers for divisibility by 3.

(i) 135 (ii) 1,471 (iii) 2,100 (iv) 3,331 (v) 31,242

3. Which of the following numbers are divisible by 5?

(i) 1,235 (ii) 5,552 (iii) 6,035 (iv) 10,001 (v) 53,550

4. Test the following numbers for divisibility by 10?

(i) 1,350 (ii) 2,225 (iii) 30,500 (iv) 13,575 (v) 20,050

5. Identify the numbers which are divisible by 5 and 10 both.

(i) 12,000 (ii) 2,145 (iii) 4,040 (iv) 12,345 (v) 7,270

6. Check the divisibility of following numbers by 2, 3, 5 and 10.

Numbers	Divisibility by 2	Divisibility by 3	Divisibility by 5	Divisibility by 10
405				
3,354				
2,340				
41,220				
34,329				

2.2 PRIME AND COMPOSITE NUMBERS

Define prime and composite numbers

Over 2000 years ago, a famous Greek Mathematician Eratosthenes, was interested in prime numbers. He arranged the numbers in 10 columns. Follow the steps he did to find out all the prime numbers between 1 and 100.



Activity

- '1' is a very special number, leave it as it is.
- Start with 2, circle it and cross out every 2nd number after it.
- Now move to 3, circle it and cross out every 3rd number after it.
- The next number that has not been circled or crossed out is 5, circle it now and cross out every 5th number.
- What is the next number after 5 that has not circled or crossed out? Circle that number and follow the same steps until you have all the numbers been crossed out or circled except 1.

The chart will look as under:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Teacher's Note

After teaching the concept of prime and composite numbers, teacher should ask the students oral questions.

The activity shows that:

All the numbers that have been circled 2, 3, 5, 7, etc. are **Prime numbers**.

All the numbers that have been crossed out 4, 6, 8, 9, 12, 14 etc. are **Composite numbers**.

Note: '1' is neither composite nor prime number.

(a) Prime numbers:

A number which has only two factors 1 and numbers itself is called a prime number.

Like 2, 3, 5 and 41 etc they can not be divided by any other number except 1 and the number itself.

There is only one prime number that is even, all the other prime numbers are odd.
What is that number?

(b) Composite numbers:

Numbers which have more than two factors are called Composite numbers.

Composite numbers can be written as a product of two or more prime numbers
 $10 = 5 \times 2$

For example: 4 can be divided by 1, 2 and 4.

18 can be divided by 1, 2, 3, 6, 9 and 18.

So, 4 and 18 are composite numbers.

Identify prime or composite numbers

Number	Factors	Conclusion
37	1 and 37	Two factors: a prime number
42	1,2,3,6,7,14,21 and 42	More than two factors: a composite number
19	1 and 19	Two factors: a prime number
77	1, 7, 11 and 77	More than two factors: a composite number

- A prime number has exactly two factors 1 and itself.
- A composite number has more than two factors.

EXERCISE 2.2

1. Identify the prime numbers from the following.

- (i) 22 (ii) 41 (iii) 63 (iv) 51 (v) 81
(vi) 119 (vii) 223 (viii) 1 (ix) 101 (x) 222

2. Identify the composite numbers from the following.

- (i) 34 (ii) 71 (iii) 163 (iv) 351 (v) 81
(vi) 19 (vii) 23 (viii) 100 (ix) 18 (x) 135

3. Separate composite and prime numbers from following.

41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60.

4. Write down all the prime numbers between 1 and 20.

5. Write down all the composite numbers between 10 and 30.

6. Write down all the prime and composite numbers between 20 and 40.

2.3 FACTORS AND MULTIPLES**List factors of a number up to 50**

1. **Factors:**

A factor divides a number completely with zero remainder.

Example 1:

Factors of 6 are 1, 2, 3 and 6

Factors of 12 are 1, 2, 3, 4, 6 and 12

The highest factor of every number is the number itself



1 is a factor of every number

Teacher's Note

Teacher should also perform some activities of factors and multiples in the class.

Example 2: Write down all the factors of

9, 10, 15, 18, 42 and 50

Factors of 10 are 1, 2, 5, and 10

Factors of 15 are 1, 3, 5 and 15

Factors of 18 are 1, 2, 3, 6, 9 and 18

Factors of 42 are 1, 2, 3, 6, 7, 14, 21 and 42

Factors of 50 are 1, 2, 5, 10, 25 and 50.



What operation do you think we use to find the factors of a number

2. Multiples:

This biscuit costs Rs 2. For every biscuit you buy, the cost will go up by addition of 2. Costs will be Rs 2, Rs 4, Rs 6 and so on. So 2, 4, 6, 8 are few multiples of 2.



Example:

Cost of 1



$$2 \times 1 = 2$$

Cost of 2



$$2 \times 2 = 4$$

Cost of 3

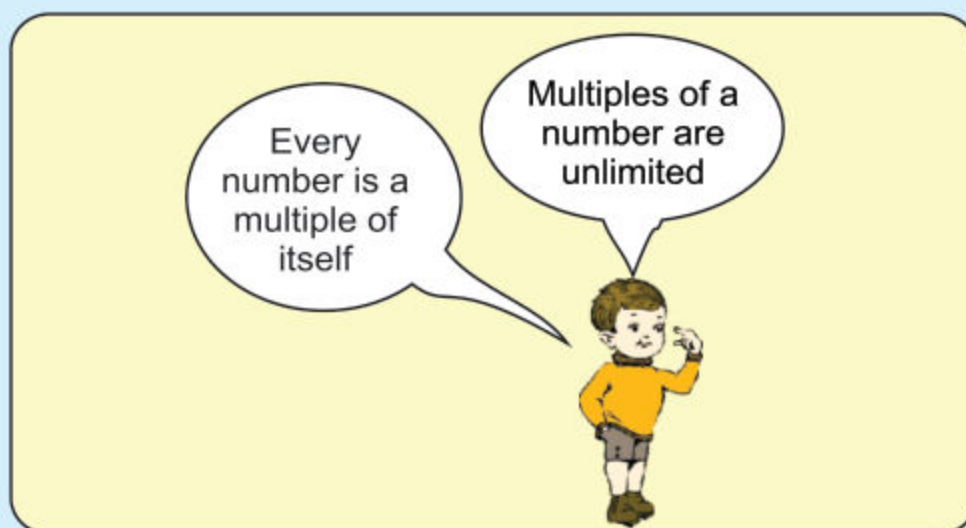


$$2 \times 3 = 6$$

So, 2, 4, 6, 8, . . . are multiples of 2.

The multiplication that you learnt in previous classes, can help to find multiples of a number. The first few multiples of 2, 3, 4 and 5 are given below:

$1 \times 2 = 2$	$1 \times 3 = 3$	$1 \times 4 = 4$	$1 \times 5 = 5$
$2 \times 2 = 4$	$2 \times 3 = 6$	$2 \times 4 = 8$	$2 \times 5 = 10$
$3 \times 2 = 6$	$3 \times 3 = 9$	$3 \times 4 = 12$	$3 \times 5 = 15$
$4 \times 2 = 8$	$4 \times 3 = 12$	$4 \times 4 = 16$	$4 \times 5 = 20$



List the first twelve multiples of a 1-digit number

Example 1: List first twelve multiples of 8 and 6.

Solution:

First twelve multiples of **8** are
8, 16, 24, 32, 40, 48, 56, 64,
72, 80, 88 and 96

First twelve multiples of **6** are
6, 12, 18, 24, 30, 36, 42, 48,
54, 60, 66 and 72

Differentiate between factors and multiples

Factors of 6 are:
1, 2, 3 and 6.
They can be counted.

$$\begin{array}{l} \boxed{1} \times 6 = \boxed{6} \\ \boxed{2} \times 3 = \boxed{6} \end{array}$$

Multiple of 6

$6 \times 1 = 6$
 $6 \times 2 = 12$
 $6 \times 3 = 18$
 $6 \times 4 = 24$
.....

EXERCISE 2.3

1. List the first twelve multiples of 4, 7 and 9.
2. Write down all the factors of 16, 26 and 45.
3. Circle all the multiples of 7 from the following numbers
14, 24, 28, 35, 45, 56, 62, 84, 69
4. Write down all the multiples of 8 between 20 and 90.
5. Write all the factors of 50 between.

2.4 PRIME FACTORIZATION**Factorize a number by using prime factors**

The process of writing numbers as a product of its prime factors is called Prime Factorization.

There are two prime factorization methods:

(1) Factor tree method

(2) Division method

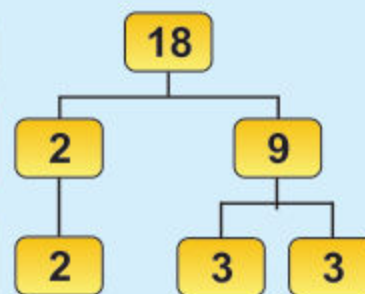
1. Factor tree method

Example 1: Factorize 18 into its prime factors.

Choose any two factors except 1 and 18

Since 9 is a composite number,
so we can write $9 = 3 \times 3$

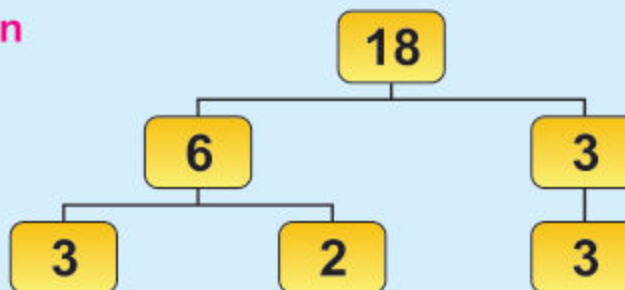
Are 2 and 3 both prime factors?
Yes, **Stop!**



Hence, we can write $18 = 2 \times 3 \times 3$

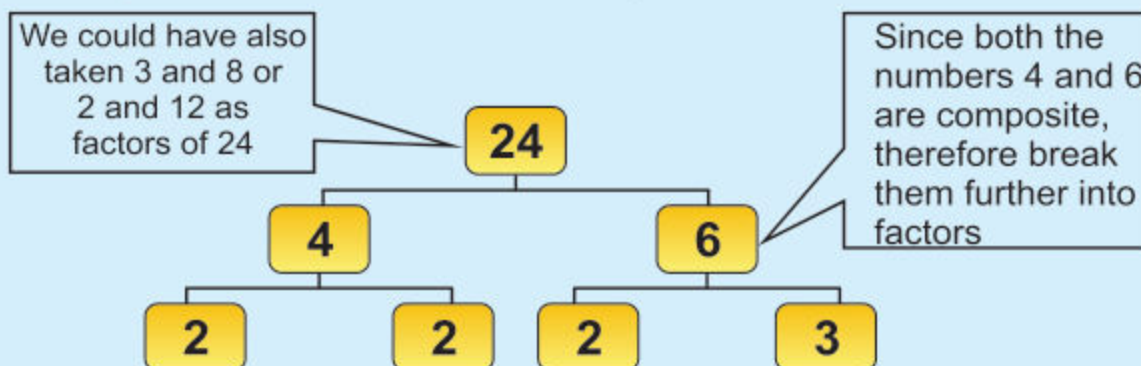
where 2 and 3 are the prime factors of 18.

**We could have also taken
3 and 6 as factors of 18**



Hence we can write
 $3 \times 2 \times 3 = 18$

Example 2: Factorize 24 into its prime factors.



Prime factors of 24 are 2, 2, 2 and 3.

2. Division method:

Example 1:

2	18	
3	9	Since, $2 \times 9 = 18$
3	3	Since, $3 \times 3 = 9$
	1	Since, $3 \times 1 = 3$

$$18 = 2 \times 3 \times 3$$

Example 2:

2	24	
2	12	Since, $2 \times 12 = 24$
2	6	Since, $2 \times 6 = 12$
3	3	Since, $2 \times 3 = 6$
	1	Since, $3 \times 1 = 3$

$$24 = 2 \times 2 \times 2 \times 3$$

EXERCISE 2.4

- Write down all the factors of the following numbers and underline the prime factors.
(i) 12 (ii) 20 (iii) 25 (iv) 44 (v) 64 (vi) 28
- Factorize the following numbers using the factor tree method.
(i) 36 (ii) 54 (iii) 27 (iv) 45 (v) 32 (vi) 82
- Factorize the following numbers using the division method.
(i) 63 (ii) 45 (iii) 72 (iv) 54 (v) 38 (vi) 90

2.5 HIGHEST COMMON FACTOR (HCF)

Determine common factors of two or more 2-digit numbers

Let us first list down the factors of any two numbers.

Example 1: Factors of 18 are: 1, 2, 3, 6, 9 and 18.

Factors of 24 are: 1, 2, 3, 4, 6, 8, 12 and 24

The common factors of 18 and 24 are 1, 2, 3 and 6 out of which '6' is the highest so, 6 is called the **Highest Common Factor**.

Find HCF of two or more 2-digit numbers using

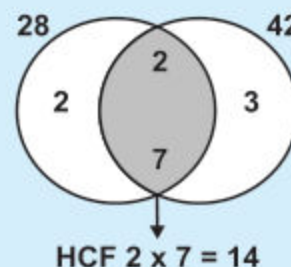
(i) Venn diagram

(ii) Prime factorization

Example 2: Find the Highest Common Factor of 28 and 42 by using Venn diagram and prime factorization method.

1. Venn diagram

Let us find the prime factors of 28 and 42 by using Venn diagram.



2. Prime Factorization

Let us find the prime factors of 28 and 42 by using division method.

$$\begin{array}{r|l} 2 & 28 \\ \hline 2 & 14 \\ \hline 7 & 7 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 42 \\ \hline 3 & 21 \\ \hline 7 & 7 \\ \hline & 1 \end{array}$$

$$\begin{aligned} 28 &= 2 \times 2 \times 7 \\ 42 &= 2 \times 3 \times 7 \end{aligned}$$

}

Loop the common factors

$$\text{HCF} = 2 \times 7 = 14$$

Product of all the common factors

Teacher's Note

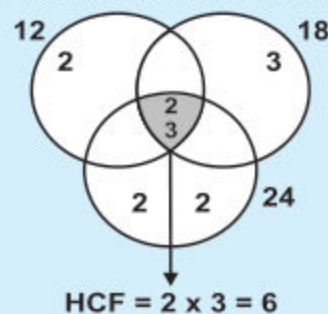
Teacher may use other set of numbers and develop the concept of finding HCF of numbers.

Unit 2 FACTORS AND MULTIPLES (Prime Factorization)

Example 3: Find the Highest Common Factor of 12, 18 and 24 by using Venn diagram and prime factorization method.

1. Venn diagram

Let us find the prime factors of 12, 18 and 24 by using Venn diagram.



2. Prime Factorization

Let us find the prime factors of 12, 18 and 24 by using division method.

2	12
2	6
3	3
	1

2	18
3	9
3	3
	1

2	24
2	12
2	6
3	3
	1

$$\begin{aligned} 12 &= 2 \times 2 \times 3 \\ 18 &= 2 \times 3 \times 3 \\ 24 &= 2 \times 2 \times 2 \times 3 \end{aligned}$$

Loop the common factors

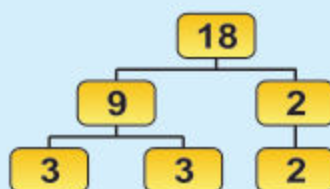
$$HCF = 2 \times 3 = 6$$

Product of all the common factors

Example 4: Find the Highest Common Factor of 12, 18 and 54 by using prime factorization method and Venn diagram.

By Prime Factorization

Let us find the prime factors of 12, 18 and 54 by using factor tree method.

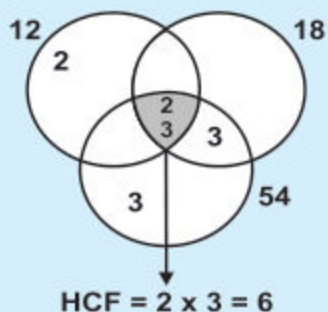


$$\begin{aligned} 12 &= 2 \times 2 \times 3 \\ 18 &= 2 \times 3 \times 3 \\ 54 &= 2 \times 3 \times 3 \times 3 \end{aligned}$$

$$HCF = 2 \times 3 = 6$$

By Venn diagram:

Let's find the prime factors of 12, 18 and 54 by using Venn diagram.



Teacher's Note

Teacher may use other set of numbers and develop the concept of finding HCF of numbers.

Example 5: Find the Highest Common Factor of 27 and 45 by using prime factorization method and Venn diagram.

By Prime Factorization

Let us find the prime factors of 27 and 45 by using factor tree method.



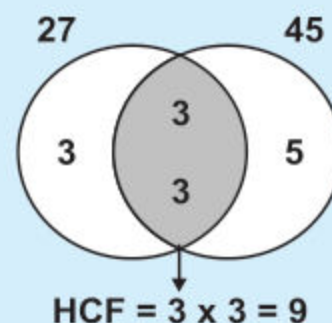
$$27 = 3 \times 3 \times 3$$

$$45 = 3 \times 3 \times 5$$

$$\text{HCF} = 3 \times 3 = 9$$

By Venn diagram:

Let's find the prime factors of 27 and 45 by using Venn diagram.



EXERCISE 2.5

- Write down all the factors of the following pairs of numbers and loop the common factors.
 - 12 and 18
 - 10 and 15
 - 22 and 44
 - 8 and 32
 - 24, 30 and 36
- Find the HCF of the following numbers by Prime Factorization Method.
 - 36 and 42
 - 28 and 42
 - 45 and 75
 - 25 and 75
 - 18 and 72
 - 32 and 64
 - 60 and 90
 - 25, 40 and 50
- Find the HCF of the following numbers by Venn Diagram.
 - 24 and 36
 - 63 and 54
 - 40 and 44
 - 48 and 84
 - 22 and 24
 - 14, 21 and 49

Solve real life problems involving HCF

Example 1: Two wires with lengths of 48 cm and 64 cm are to be cut into pieces of same length. Find the greatest possible length of the pieces.

Solution: We have to find the greatest of length in which each of wire can be divided. In order to find the greatest possible length of the pieces of each wire, we'll have to find the HCF of 48 and 64.

2	48
2	24
2	12
2	6
3	3
	1

2	64
2	32
2	16
2	8
2	4
2	2
	1

$$48 = 2 \times 2 \times 2 \times 2 \times 3$$

$$64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$\text{So, HCF} = 2 \times 2 \times 2 \times 2 = 16$$

Hence, the greatest possible length of the piece of wire should be 16 cm in which each wire could be cut equally.

Example 2: There are 3 companies of 80, 112 and 144 scouts in a school. Find the highest number of scouts in which each team can be distributed equally.

2	80
2	40
2	20
2	10
5	5
	1

2	112
2	56
2	28
2	14
7	7
	1

2	144
2	72
2	36
2	18
3	9
3	3
	1

$$\text{Therefore, } 80 = 2 \times 2 \times 2 \times 2 \times 5$$

$$112 = 2 \times 2 \times 2 \times 2 \times 7$$

$$144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$$

$$\text{So, HCF} = 2 \times 2 \times 2 \times 2 = 16$$

Teacher's Note

Teacher may take other examples related to real life i.e number of hours spent in ploughing field, distance covered by the car, cost of crops produced by farmers, things sold by shopkeeper and clear the use of finding HCF.

Example 3: Sheeba has three pieces of cloth. The measure of three pieces are 12, 24 and 30 meters respectively. She needs to cut the pieces into of equal width that are as wide as possible.

we can find width by using HCF, now

$$\begin{aligned} \text{Therefore, } 12 &= 2 \times 2 \times 3 \\ 24 &= 2 \times 2 \times 2 \times 3 \\ 30 &= 2 \times 3 \times 5 \end{aligned}$$

$$\begin{array}{r|l} 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 24 \\ \hline 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 30 \\ \hline 3 & 15 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$\text{So, HCF} = 2 \times 3 = 6$$

She should cut each piece to be 6 meters.

EXERCISE 2.6

1. What is the maximum number of students among which Haider can distribute 36 sweets and 48 jelly to divide them equally?
2. Find the greatest number of bundles of books in which 18 Science books, 24 Maths books and 36 English books can be arranged equally.
3. Two ribbons with the length of 44 m and 66 m are to be cut into pieces of same length. Find the greatest length of the piece of ribbon?
4. Mona has two pieces of tablecloth, one is 48 cm wide while the other is 90 cm wide. She wants to cut both pieces into strips of equal width. What should be the maximum width of each strip?
5. A class teacher has 32 red balloons, 28 blue balloons and 20 green balloons. She wants to distribute balloons equally among the students. Find the maximum number of balloons which could be distribute between students.
6. Zulfiqar has three containers containing 144, 176 and 256 litres of coconut oil respectively. Find the capacity of the largest tin by which he can measure the oil exactly.